

Domain Controller and Active Directory Lab

Part 6 – Establishing Dynamic Host Configuration Protocol

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Date: August 19, 2026

Technologies Used

- **Oracle VirtualBox**
- **Windows Server 2022**
- **Server Manager**
- **Active Directory Domain Services**

OBJECTIVE:

The objective of this lab is to establish the Dynamic Host Configuration Protocol (DHCP) for the DC. Establishment of the DHCP allows future clients to have addresses assigned to them from a configured address range on DC1.

Lab Setup:

This lab continues from Part 5 where NAT and RAS were created, allowing clients to access external internet through DC.

For this lab will establish Dynamic Host Configuration Protocol on DC, which will lease IP addresses (ranging from 172.24.8.100 – 172.24.8.200) for future clients that request address from DC.

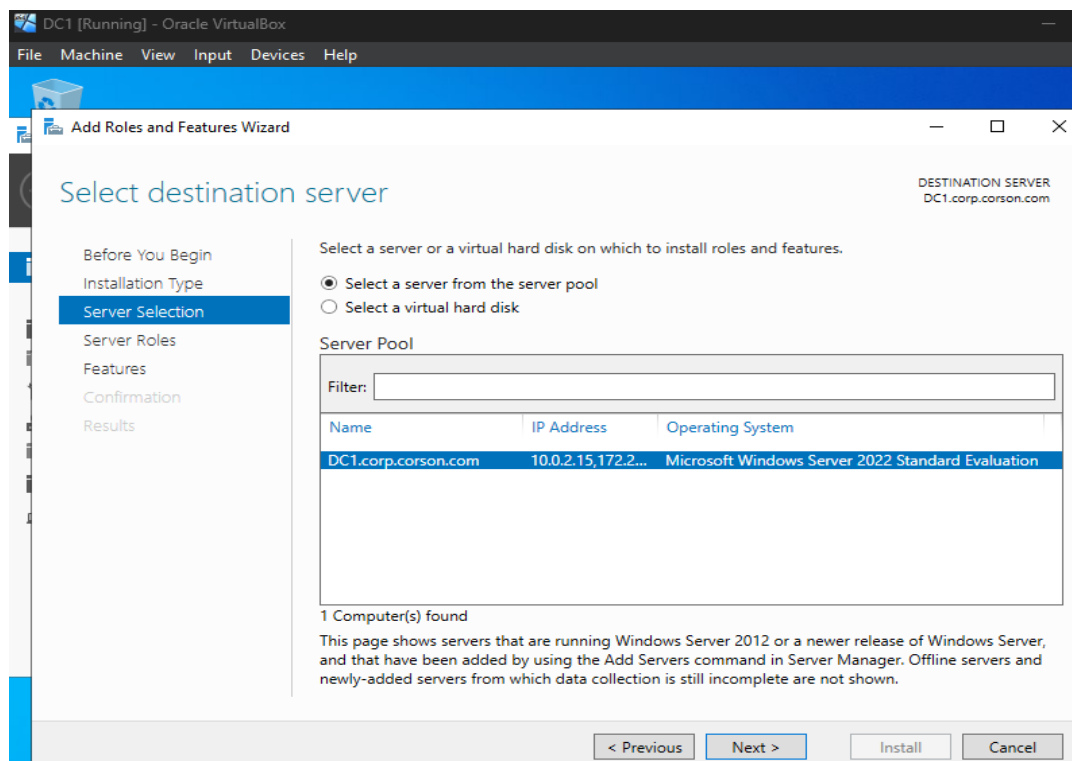


Figure 1. Using Server Manager, DC1.corp.corson.com was selected as the destination server where the DHCP Server role will be installed.

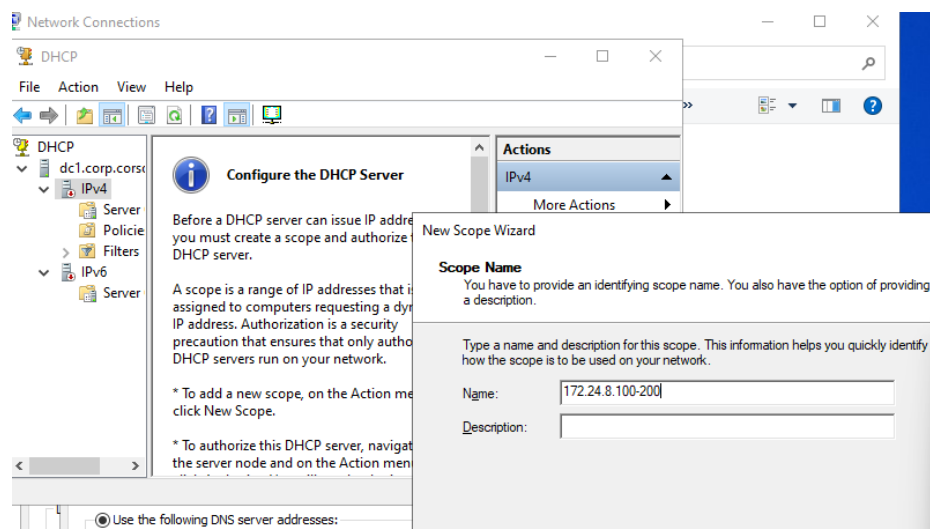


Figure 2. Creating and naming a new DHCP scope that will be used to define the range of IP addresses available for lease to future clients on the internal network.

New Scope Wizard

IP Address Range

You define the scope address range by identifying a set of consecutive IP addresses.

Configuration settings for DHCP Server

Enter the range of addresses that the scope distributes.

Start IP address: 172 . 24 . 8 . 100

End IP address: 172 . 24 . 8 . 200

Configuration settings that propagate to DHCP Client

Length: 24

Subnet mask: 255 . 255 . 255 . 0

Figure 3. Defining the DHCP scope address range from 172.24.8.100 to 172.24.8.200. These addresses will be available for the DHCP server to lease to future clients connected to the internal network.

New Scope Wizard

Lease Duration

The lease duration specifies how long a client can use an IP address from this scope.

Lease durations should typically be equal to the average time the computer is connected to the same physical network. For mobile networks that consist mainly of portable computers or dial-up clients, shorter lease durations can be useful. Likewise, for a stable network that consists mainly of desktop computers at fixed locations, longer lease durations are more appropriate.

Set the duration for scope leases when distributed by this server.

Limited to:

Days: 8 Hours: 0 Minutes: 0

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Figure 4. Configuring the DHCP lease duration to 8 days, determining how long an assigned IP address can be leased to a client before the lease must be renewed.

More Actions ▶

New Scope Wizard

Router (Default Gateway)

You can specify the routers, or default gateways, to be distributed by this scope.

To add an IP address for a router used by clients, enter the address below.

IP address:

172 . 24 . 8 . 1	Add
	Remove
	Up
	Down

Figure 5. Configuring the Domain Controller's static IP address (172.24.8.1) as the default gateway, allowing future clients to route traffic through DC1 to access external networks.

New Scope Wizard

Domain Name and DNS Servers

The Domain Name System (DNS) maps and translates domain names used by clients on your network.

You can specify the parent domain you want the client computers on your network to use for DNS name resolution.

Parent domain: corp.corson.com

To configure scope clients to use DNS servers on your network, enter the IP addresses for the servers.

Server name:	IP address:	
		Add
	172.24.8.1	Remove
		Up
		Down

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Figure 6. Configuring the parent domain as corp.corson.com and assigning DC1 (172.24.8.1) as the DNS server that will be provided to future DHCP clients.

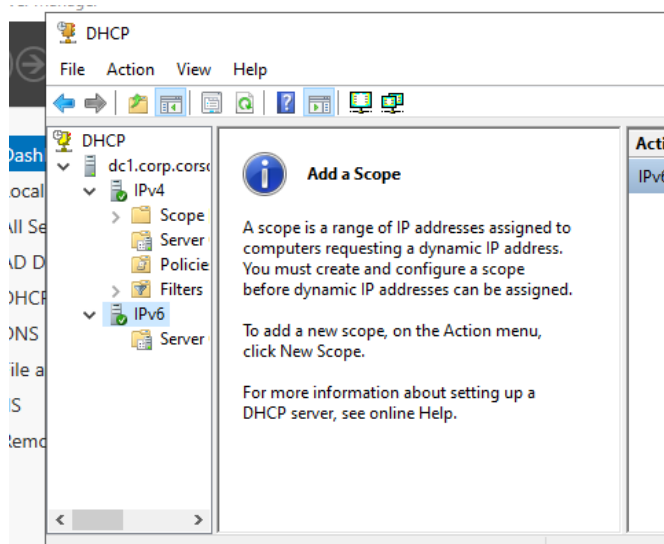


Figure 7. Verifying that the newly configured DHCP scope is active. The green status indicators show that the DHCP server and IPv4 scope are operational and ready to provide network configuration to future clients.

What Changed?

Before this configuration, DC1 did not have a method of automatically assigning IP addresses to future clients on the internal network. Clients would have required their network settings to be manually configured.

After installing and configuring DHCP, DC1 can now lease IP addresses to clients from 172.24.8.100-172.24.8.200 scope. Clients receiving a DHCP lease will also be provided 172.24.8.1 as their default gateway and DNS server, along with the corp.corson.com parent domain.

This prepares the internal network for future clients to automatically receive the network configuration needed to communicate with DC1 and access external networks through the previously configured RRAS/NAT service.